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Human resource development and needs analysis for nuclear power plant deployment in Nigeria

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A B S T R A C T

The fulcrum of economic development is a sustainable supply of electricity. Nigeria is plagued with blackouts, with one of the lowest per capita electricity consumption in the world (circa. 120 kWh per capita). Hence, policies have been instigated to integrate electricity generation from nuclear power plants (NPP) on or before 2027. However, a critical requirement for NPP generation is the implementation of robust human resource development (HRD) programs. This paper presents the perspective of Nigeria in assessing human resources needs over the entire NPP lifecycle following the milestone approach and employing the IAEA's Nuclear Power Human Resource (NPHR) modeling tool. Three workforce organizations are in focus including the owner/operator, regulators, and construction workers following three decades timeframe (2015–2045). The results indicate that for the study period, a maximum of approximately 9045 personnel (73% construction workers, 24% owner/operator, and 3% regulators) should be directly involved in the NPP program just before the commissioning of the third NPP in 2033. However, this number decreases by about 73% (2465 personnel including 94% operator and 6% regulator) at the end of the study timeframe. The results can potentially provide clarity and guidance in HRD decision-making programs.

1. Introduction

In the past few decades, there has been an unprecedented increase in the demand for energy globally, which to some extent is the precursor to the fluctuating climatic conditions experienced around the world. However, in 2015, because of the changing climate, over 190 countries signed the Paris Agreement that stipulates that one of the potential solutions to curbing climate change is to switch to alternative energy sources such as nuclear and renewable energy [1]. Consequently, Nigeria, which is the subject of study, has made strides to incorporate nuclear power into its electricity generation mix within the next two decades [2].

However, Nigeria is the most populous black nation in the world with an estimated population of over 200 million as of April 30, 2019 [3]. Despite the increase in its population from its independence in 1960 to date, there has been no commensurate investment in the power sector to meet the rising energy demand. Similarly, Nigeria's current installed power generation capacity of 12 522 MW is unable to sustain the demands of the country [4]. A reality check indicates that the actual power generation has been fluctuating between 1500 and 4000 MW in the past ten years [4], which is equivalent to an average of 120 kWh per capita. Accordingly, power outages are frequent, and self-generation is a necessity for the industry and other consumers. Hence, another reason why Nigeria has mandated the introduction of nuclear power into its energy mix.

Consequently, nuclear energy has been touted as a possible source of power generation in Nigeria, not only to alleviate its current dire electricity needs but also to mitigate climate change. Conversely, a quick historical outlook indicates that although Nigeria conceived its nuclear power program in 1976 through promulgating an Act to establish the Nigeria Atomic Energy

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Commission (NAEC) (the focal organization responsible for the promotion of nuclear or atomic energy in all its ramifications in Nigeria), nothing was done until its reactivation 30 years (2006) later. Since then, NAEC has ramped up efforts to ensure that Nigeria builds its nuclear power plant (NPP). Accordingly, Nigeria has signed three (3) Inter-Governmental Agreements (IGAs) with the Russian Federation, including Cooperation on the Development of Nuclear Energy for Peaceful Uses (signed 2009), Cooperation in the Design, Construction, and Decommissioning of Nuclear Power Plant(s) in the Federal Republic of Nigeria (signed 2012), and Cooperation in the Construction of a Multi-Purpose Research Reactor (MPRR) Complex (signed 2016). Furthermore, two Project Development Agreements (PDAs) signed on October 30, 2017 with timelines to implement the IGAs have been put forward to construct four nuclear power plants (proposed total capacity of 4000 MW) to be tentatively delivered by 2037 [5].

In view of that, a country's decision to embark on a nuclear power program is a major task with very high demands. It will take more than a decade of preparation and a long-term commitment of about a century. The country must have in place adequate infrastructure that ensures that the nuclear power plants are used in a safe, secure, and peaceful manner [6]. The International Atomic Energy Agency (IAEA) in assisting member states to meet this requirement developed a milestones approach (see Table 1). This approach has been internationally accepted for implementing nuclear power programs, and the method breaks the nuclear power program into three phases that serve as a comprehensive guide to help member states understand the requirements for the peaceful, safe, and secure use of nuclear power. Moreover, the procedures taken and activities included in the NPP project are divided into three phases within the milestone approach. A milestone is achieved after each phase is completed. The IAEA [7] has therefore outlined three (3) distinct phases and three (3) milestones for clarity.

The IAEA milestone approach further stipulates nineteen (19) key infrastructure issues (as shown in Fig. 1) that should be addressed before a nuclear power plant (NPP) may be introduced into a country's energy mix [7].

Human resource development (HRD) is one of the critical infrastructure requirements that must be implemented before the construction and operation of the NPP. HRD is particularly challenging to implement for developing countries because of the specialized workforce requirements for nuclear reactors [8]. Many of the required skills are not readily available in these countries and usually require over 10 years to be developed [6]. NPPs require a highly technical and competent workforce for their functionality and sustainability. Hence, adequate planning for HRD is expedient because the NPP project construction and operation can span 80 years. HRD planning involves evaluating the current capacity of the national workforce, identifying gaps in meeting required skills, and developing strategies to close these gaps [6]. Moreover, the IAEA, in its quest to assist member states in developing robust and sustainable HRD frameworks for their nuclear power program workforce planning, has acquired the Nuclear Power Human Resources (NPHR) modeling tool. The tool will assist member states, such as Nigeria, to evaluate the human resource requirements for the key organizations of the nuclear program which will cover its life cycle. Hence, the objective of this paper is to assess the full range of Nigeria's NPP workforce requirements 2015 to 2045 using the NPHR modeling tool. The 2015 date is proposed because it was when Nigeria attained milestone 2 of the IAEA's milestone approach to nuclear power plant(s) deployment.

The subsequent sections of this paper include the following: Section 2 is the literature review which involves a brief discussion of the HRD analysis carried out in other industries and countries. Section 3 puts forward the methodology consisting of a short description of the NPHR modeling tool and assumptions made for its implementation. Section 4 illustrates the application of the NPHR model to the case study of Nigeria's nascent Nuclear Power Program. Lastly, Section 5 presents the conclusion and recommendations for future work. The novelty of this work is the integration of Nigeria's data with the IAEA milestone approach in terms of NP planning, and the first detailed presentation which demonstrates the robustness and effectiveness of the NPHR modeling tool.

2. Review of literature

In this section, an overview of the different HRD analysis techniques used in other industries is put forward. This is closely followed by a short discussion of HRD analysis in nuclear power plants using other HRD tools. The section concludes with a short discussion of the utilization of the NPHR modeling tool so far.

2.1. HRD analysis in other industries

Different techniques have been proposed and/or developed over time for HRD analysis. However, these techniques adopted in several industries for forecasting HRD are generally classified into two broad categories including quantitative and qualitative analysis approaches. While the quantitative analysis approach involves the use of different statistical methods for predicting HRD requirements such as trend analysis, ratio analysis, and regression analysis, the qualitative analysis approach involves drawing inferences from experienced managers (also called expert judgments). Some examples of qualitative approaches are the Delphi method, nominal group technique, and scenarios creation method [9].

Predicting workforce needs in human resource management

Table 1
IAEA milestone approach.

	Phase 1	**Milestone 1**	**Phase 2**	**Milestone 2**	**Phase 3**	**Milestone 3**
Team Activities	Pre-project activities Considerations before a decision to launch a nuclear power program is taken	Ready to make a knowledgeable commitment to a nuclear power program	Project development Preparatory work for the contracting and construction of a nuclear power plant after a policy decision has been taken	Ready to invite bids/negotiate a contract for the first nuclear power plant;	Final investment Activities to implement the first nuclear power plant	Ready to commission and operate the first nuclear power plant.
Time Frame	**At least 10–15years**					

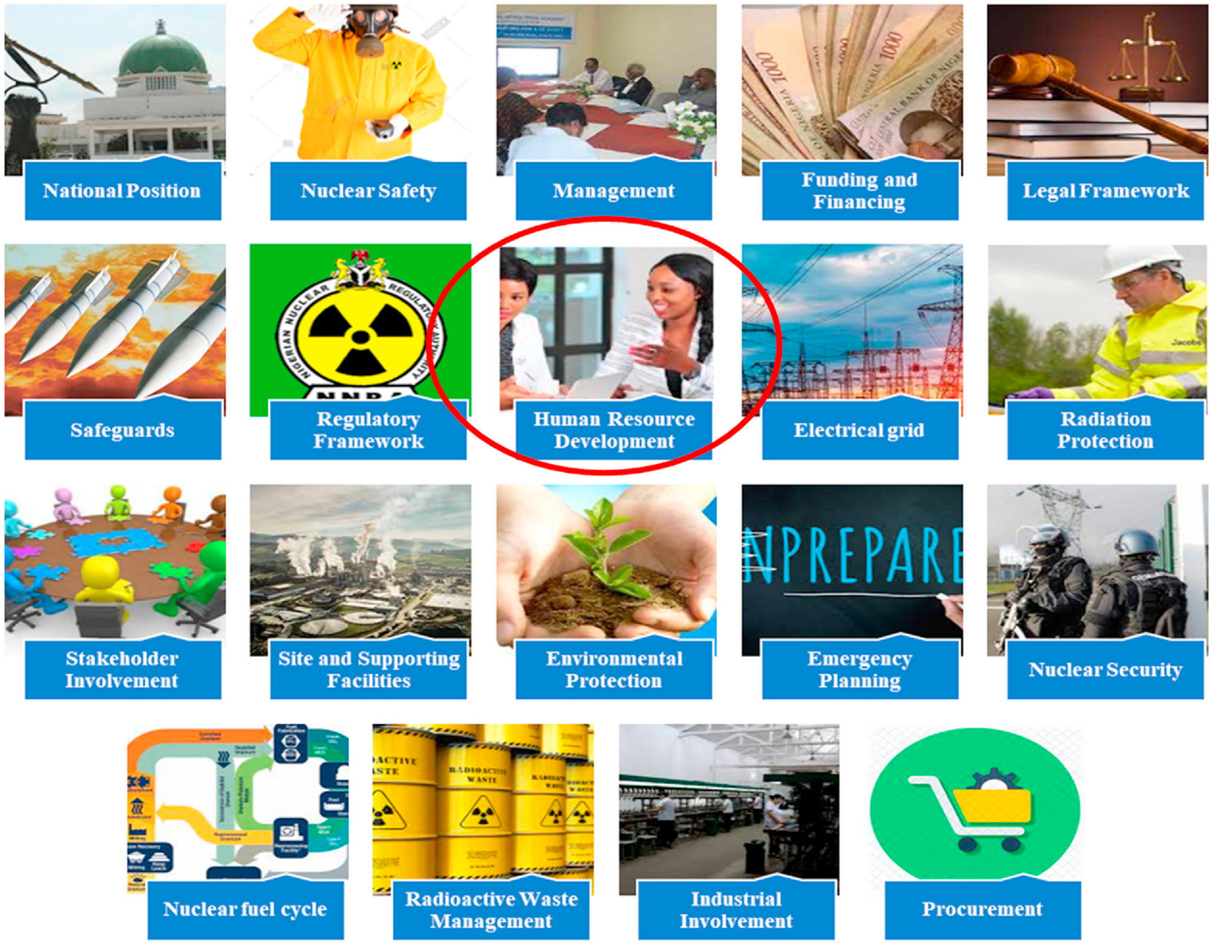

Fig. 1. Nineteen (19) key infrastructure issues.

has gone a long way in helping industries plan properly, reduce cost, and eliminate wastage and downtimes. In a recent review, Samwel [10] identifies human resource planning (HRP) as one of the most important practices carried out in an organization. In the study, the relationship between effective human resource planning and achieving organizational needs with the importance of human resource planning in meeting organizational manpower needs were highlighted. The author concludes that human resource planning helps an organization achieve its results more effectively and efficiently.

Nonetheless, for effective human resource planning, various tools have been developed in the HRD space including Human Resource Predictive Analytics (HRPA), which uses a quantitative approach in human resources analysis based on data collection and predictive tools. For instance, the HRPA was used in the US wind turbine industry, whereby results showed that women employees performed 5% better than men. Consequently, the wind turbine industry improved their recruitment policies [9] to reflect the results. In another study, the RAFAELA system, which is an HRD modeling tool developed in Finland for nurses' human resource management, is also widely applied in other parts of Europe for nurses' workforce assessment. This system is based on the quantitative approach measures nursing intensity (NI) and allocation of nursing staff. The system also has the further merit of assisting in classifying patients' needs to provide a proper allocation of nursing care. It is noteworthy that the RAFAELA system works on three levels of data abstraction, thereby assisting nursing managers in forecasting and ensuring quality care [11]. In a more recent study to predict adequate manpower requirements for an academic institution (Federal University of Technology Akure, Nigeria), a regression analysis-based model is employed. The methodology involves the collection of data on students and staff over a 13-year period. The regression-based model is used to analyze and forecast the institution's manpower requirements as the students' enrolment number increases [12]. The workforce structure of the University of Uyo, Nigeria, is analyzed in another research using the Markov chain predictive model. The model is used to identify the current structure of the university, identifying the capacity gaps, and predicting the academic requirements over the next five years [13]. The dynamic human resource predictive model is another workforce planning tool that is constructed based on Material Requirements Planning (MRP) [14]. The MRP is a planning tool that has been successfully used for inventory planning in many organizations. The idea behind MRP is the application of a predictive model that uses a non-linear regression model to forecast workforce demand based on existing workforce data. For example, it is applied to a government agency using 11-year period data to accurately predict the workforce requirements measured using a Mean Absolute Percentage Error (MAPE) of below 5%. Furthermore, an estimate of the manpower flow for the Nigeria Port Authority (NPA) is evaluated using multivariate linear regression. The model uses data for 20 years of manpower to analyze the current situation of the NPA workforce and can forecast the required yearly manpower flow [15].

The manpower requirements for other industries have also been estimated using different HRP approaches. For instance, the Markov Chains Model and Job Coefficient are used to predict the supply and demand of human resources in the nuclear industry. The approach identified and predicted the human resource gap in the nuclear industry between 2008 and 2030 employing data from surveys and nuclear energy yearbooks of about 17 years [16]. Moreover, in the steel manufacturing industry, Alam et al. [17], employed autoregressive integrated moving average (ARIMA) and fuzzy associative memory (FAM) neural network model to estimate the manpower requirements for mechanical engineering of the steel manufacturing industry group in the state of West Bengal, India. The results of the study predicted a downward manpower trend in the supply of engineers, mostly due to the modernization of various factory units, hence, making the engineering manpower less relevant. Consequently, the authors conclude that the two approaches (ARIMA and FAM) can be used in workforce prediction with minimal error and the results can assist in forewarning management to find sustainable solutions to the predicted downward manpower trend. In the construction industry, Ho [18] developed a model for workforce planning using a single-variable first-order gray model. The approach was able to predict construction manpower requirements over the next three months horizon. However, the model required data obtained from the first 3 months of 1992 to the last three months of 2007 to make predictions showing minimal errors.

The conclusions drawn from the reviewed literature are: (i) human resource planning is vital for organizations to perform effectively and optimally, (ii) human resource planning cuts across different organizations and fields (industries, academia, health and etc.), (iii) different approaches are employed for human resource planning in different sectors and organizations, (iv) the approaches used require in-depth understanding of the organization and its system of operation, (v) historical data is important, (vi) there is the need to identify and understand the modeling tool and ensure that it incorporates all the unique features of an organization. This paper puts into consideration the need for a model to meet the aforementioned requirements. Hence, the model should be capable of workforce planning tailored to a developing country and newcomer to the nuclear power plant sector such as Nigeria. It is important that the approach for human resource planning overcomes the limitation of inexperience, unavailable data, and manpower challenges in developing countries. Therefore, the NPHR model is employed for this study because it is designed for NPP based on a proper understanding of the complexities involved in NPP. It also considers the duration of operation of NPP as well as data from other nuclear power plants from other member states.

2.2. HRD analysis for nuclear power plants using other HRD tools

Human resource planning in the nuclear industry is of critical concern because of the complexity and lifespan of NPPs. In Japan, a trend analysis approach is used to investigate the manpower requirements of the nuclear industry. This analysis was done as part of the overall HRD plan created to strengthen the nuclear program after the challenging experience of the Fukushima earthquake [19].

In another case, the Armenian nuclear industry planned the addition of electricity to its grid by building an additional nuclear plant. However, the plan was later postponed due to the life-extension of the existing NPP. When the plan for the new unit was in force, the country was already using nuclear power for base load while the plans for the construction of the additional plant were treated as a new experience because of the long period between the previous plant and the planned construction. In this respect, the human resource analysis for the Armenian proposed new build NPP is developed based on expert judgment [20]. Though the Armenian plans for construction are currently suspended, their workforce plan approach is still a lesson for other countries to learn from.

Bangladesh, a newcomer nation to the nuclear power industry already has a nuclear power plant presently under construction (i.e., the Bangladesh Rooppur NPP (RNPP) construction started in November 2017). For the RNPP human resource planning, a Joint Training Advisory Commission (JTAC) involving Bangladesh, India, and Russia was constituted. The JTAC in their HRD plan (which was based on expert judgment) projected that a workforce of about 7000–8000 craftsmen will be required during construction while about 2700 personnel will be needed during the operations of the NPP. Based on this workforce prediction, the JTAC was able to draw up and implement strategies to meet their manpower requirements [21].

The United Arab Emirates (UAE) commenced the construction of its first NPP in July 2012 and connected the NPP to its grid in August 2020. To cater for the development of the UAE HRD planning, collaboration among the Emirates Energy Commission (ENEC), the Federal Authority for Nuclear Regulation (FANR), and their Nuclear Energy Planning Implementing Organization (NEPIO) was constituted. These three organizations conducted internal reviews of the projected workforce for its NPP. The results stipulated that around 1000–1500 preoperational personnel will be needed in the year 2016, while over 2000 operational personnel should be on ground in the year 2020 during the commissioning of its first plant [17]. To meet up with the workforce requirements, the UAE managed their nuclear power program based on contractor services [22]. They developed a long-term plan of training ingenious experts for managing their NPP. For the short-term plans, they deployed experienced expatriates for the construction and management of the NPP which will then transit to local experts over time [23]. The UAE's action in implementing its nuclear power program is commendable and worthy of emulation by other newcomer countries. This proves that with adequate planning, the process of building a nuclear power plant can be achieved faster than the usual slow and sequential development process.

From the literature reviewed, it is evident that the HRD approach adopted largely depends on the peculiar situation of a country. The result of the workforce plan from the nuclear plant programs of the countries considered further shows that projections of workforce needs are affected by certain factors such as the country's population, present educational and training infrastructures, technology, size of NPP, and language. These factors are considered in the NPHR and therefore bolster the advantage of the tool as the subject of interest for this study.

2.3. NPHR use in other countries

In the last 10 years, the IAEA has conducted training courses on the use of the NPHR modeling tool in 20-member states. However, to the best of our knowledge and to date, only three countries including, Turkey, Kenya, and Morocco have practically explored the NPHR modeling tool to plan their HRD requirements [24].

For instance, in the Turkey country case study, Gurgen and Ergun [25] presented a sample application of the NPHR modeling tool to determine the number and type of workforce required in only the NEPIO since not much information was available to determine the workforce requirements in the owner/operator organization, regulatory body, and the construction organizations. Furthermore, the results indicate that for the two PDAs (i.e., the Akuuyu and Sinop project) 140 personnel should be engaged in the NEPIO by the year 2021. Moreover, it is worth stating that the Akuuyu and ATMEA projects represent four units each of the VVER-

1200 and ATMEA reactor technologies respectively. However, the results pose a great limitation and should not be used as an actual representation of the Turkey NPP since most of the data is derived from IAEA documents and the USA.

In the case of Morocco [24], three nuclear key organizations (NKO) were identified as vanguard participants in the realization of their NPP, analogous to the NEPIO, potential owner-operator, and regulatory authority. In the utilization of the NPHR modeling tool, considering the peculiarities of Morocco, it was assumed that four units of at least 1000 MW should be constructed on or before the year 2043, with an NPP starting date of 2020. The results indicate that a peak number of 50 workers will be required in the NEPIO during the preconstruction phase. Also, when the regulatory authority is considered, a maximum number of 215 personnel are required during the construction phase but tapers down to 156 workers during the full nuclear power plant operation. Furthermore, considering the owner-operator organization, 5630 staff (peak value) should be engaged during the construction phase while there is an approximate 60% drop (i.e., 2190 personnel) in workforce requirements during the NPP operation phase.

In Kenya, the Nuclear Power and Energy Agency (NuPEA) is responsible for the development of the nuclear power HRD. However, Kenya as a newcomer nation has only applied the NPHR modeling tool to simulate the early phases of development, construction, and start of the NPP. Hence, the results are yet to be published and are not readily available as an open source.

In conclusion, this section has provided an overview of HRD planning in the nuclear industry as well as other industries with a focus on the NPHR modeling tool. In the next section, a brief description of the NPHR modeling tool is presented and the critical assumptions employed for its utilization in HRD.

3. Methodology and description of the NPHR tool

The NPHR model is a tool provided by the IAEA to assist member states in assessing their plans for human resource development and the key prerequisites for new NPP [24]. The NPHR modeling tool may be used to assess the workforce needed for the NPP during its life cycle, from making the decision to construct nuclear power plants to the decommissioning of such plants (i.e., NPP life cycle). The tool also has the advantage of being amenable to expanding NPP and to those NPP requiring an examination of a restart or an extension of life. The NPHR modeling tool, therefore, extrapolates the critical workforce requirements at each stage of the NPP life cycle using the IAEA milestone approach [7]. It can also be utilized to probe or assess the staffing models for such organizations as the NEPIO, regulatory body, owner/operator, and construction contractors who are critical contributors to the NPP. Moreover, the NPHR model has the additional feature of appraising a country's education and training strategies, while accounting for associated industries that may supply or compete for the workforce [26].

The NPHR modeling tool functions on a set of customized data. These data reflect the life cycle of a nuclear power program thereby enabling the users to understand the subtleties of the workforce requirements. The tool includes open-source data on the staff of nuclear power plants and workers required during construction, which can fluctuate to a large extent in terms of number and type of workers required. The model further provides options for the user(s) to select different approaches to staffing while investigating the implications on the national workforce to support the nuclear power program. As countries develop their NPP, they are required to customize their datasheets to reflect more accurately their national position. Hence, the set of skills needed will vary from country to country or region to region and depend on the prevalent national power and regulatory setup as well as the technology design type utilized. It is thus evident that strategic planning arrangement(s) need to be implemented to ensure that fundamental education and training programs are in place to provide qualified personnel.

3.1. Technical details

The NPHR modeling tool is constructed on a commercial systems' modeling platform called Stella, marketed by isee systems inc. (www.iseesystems.com) [27]. A license is required to utilize the NPHR modeling tool on Stella, while Excel format data files are provided to customize the data to a representative country or region. The Stella software uses flow and stock modeling elements, also referred to as stock-flow diagrams (SFDs) which allow model construction to portray the transition of staff from the education and training programs into the nuclear workforce or industry. Moreover, stocks in SFDs (see Fig. 2) refer to elements that amass or collect materials and vary according to the materials flowing in and out of it [28]. On the other hand, flows are the elements of system dynamics which ensure the movement of materials in and out of stocks (see Fig.). Another critical element of SFDs is converters used to represent parameters, equations, and constants, used to influence rates of flow of materials. A major advantage of using system dynamics platforms such as Stella Architect is its ability to model continuous flow and feedback loops using connectors.

The required parameters used to portray the movement of staff can easily be entered into the Excel format data file (supplementary data) to customize the model for any nation. The NPHR model also allows users to include alterations, which can be used to include other parameters, such as the cost implications of workforce introduction to NPP. The model can also communicate with external models and programs (such as Microsoft Excel as

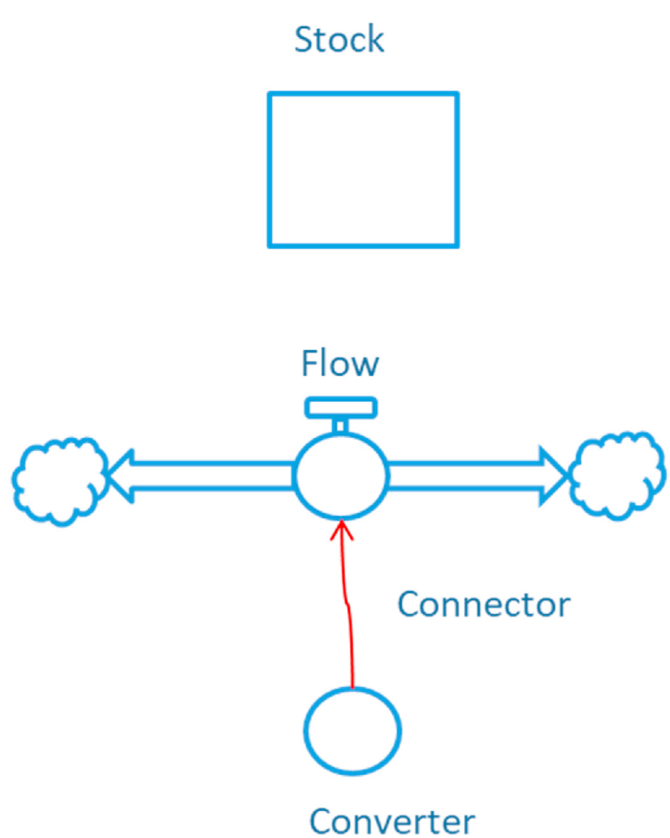

Fig. 2. Main elements of the Stock-flow diagram main.

discussed earlier) for variable input and alternative visualization of results.

3.2. Key assumptions

A PWR typical plant life of 60 years is considered with the option of a 20-year license extension. Furthermore, the NPP workforce is derived from Nigeria's 2019 population of about 200 million growing at an annual estimated rate of 2.64% [3]. The Build-Own-Operate-Transfer (BOOT) vendor contracting option is assumed whereby the vendor has maximum free will to recruit its personnel for the program. However, an option for the vendor and country to provide an equal split of the workforce may also be put forward. The BOOT option also accounts for the time taken for the vendor organization to transition to the owner-operator organization (i.e., 10 years). A current energy demand of 24.37 GWh [29] in Nigeria is placed as the starting point for the model analysis. It is further assumed that Phase 2 of the milestone approach ends in early 2022 while phase 3 which includes the construction of the NPP ends as the first 1000 MW units are commissioned in the third quarter of 2027. It is anticipated that the construction of each PWR plant with 2 units will take about 6.5 years with a license review period of at least 3.5 years [30]. Since the data for the entire national workforce in Nigeria is not readily available, a scaled value is utilized for certain skills consistent with the USA's 2019 population workforce. However, such specialized skills as boilermakers or millwrights are not scaled in this study since they are to the best of our knowledge not available in Nigeria. Hence, it is assumed that the vendor would provide the expertise to cover such skill gaps. Table 2 below displays a summary of the key assumptions. The initial workforce age distribution used in the model, mirrors NAEC's 2020 workforce age distribution shown in Fig. 3.

It is indicative that the workforce distribution, as portrayed in Fig. 3, will change over time due to worker(s) leaving the organization or industry. However, the term attrition has been used to describe the process of leaving an organization or industry and it includes the following three critical bases: job change, termination, and retirement. Attrition in this study is illustrated in Fig. 4, where it is assumed that job change is more prevalent for workers less than 30 years old while retirement is a major contributor to loss of workforce for people over 60 years old. It is worth stating that the official age for retirement from the public service is 60 years in Nigeria. However, because of the peculiarity of the nuclear power program, the retirement age is extended by five years. Notwithstanding, to ensure a sustained supply of workforce to meet the NPP requirements, the recruiting strategy exploited indicates that 60% of workers should derive from national education programs. A further 10% apiece should come from the Technical Support Organizations (TSO) and Related Industries respectively, while the remaining workforce should be experienced hires.

It is further assumed that the regulatory workforce consists of 94% professionals, 2% management, and 4% semi-skilled workers determined by equation (1) below

$$\text{Technical staff} = 1.24 * (33 + 18 * \text{operating reactors} + 41 \text{ licenses}) * 1.24 \tag{1}$$

In Eq. (1), “33” shows the base capability and “1.24” accounts for waste and other users. The reader is directed to the supplementary material which shows the detailed split of the regulatory workforce. However, in the results section, the regulatory body staff is divided into six functional areas including technical, safeguards, legal, scientists, security, and others [33]. The reader is referred to the supplementary that provides detailed information of the data inputs and references.

Following the short description of the NPHR modeling tool and the key assumptions put forward in this section, a case study of the application of the NPHR modeling tool to Nigeria is described in section 4.

4. Case study

This section is divided into two parts including section 4.1 that illustrates a base case scenario whereby the overall workforce requirements for Nigeria in the timeframe (2015–2045) are portrayed and section 4.2 which describes the specific skill sets needed by Nigeria, hereafter referred to as the domestic country over the same time frame.

4.1. Base case scenario

The NPHR modeling tool formulated on the Stella and *iThink* Version 1.3.1 consists of 992 interacting variables, 116 stocks, 169 flows, and 707 converters. The model also comprises 226 constants, 650 equations, and 650 graphical functions (i.e., time-varying functions whereby a different value is allotted for every model time step). An 8 GB RAM personal computer with Intel® Core™ i5-4210U processor @ 1.70 GHz is used, and the model solves in a few seconds.

The results obtained from the NPHR tool indicate that for the construction staff portrayed in Fig. 6a, the total number of workers involved (vendor and domestic staff) peaks at approximately 6678 personnel during the construction of the third unit (i.e., the third quarter of 2032) whereby it is assumed about 27% of the workforce

Table 2
NPHR key assumptions.

Item	Description/Quantity
Industrial objectives	Electricity generation.
Population	200 963 599
Population growth rate	2.64%
Reactor technology	Light water reactors (LWRs) with VVER reactor type taken as an example since Rosatom is Nigeria's potential vendor
Number of NPP projects	At least one nuclear power plant project either at Geregu site, Kogi State or Itu site in Akwa Ibom, Nigeria (see Fig. 5)
Unit size	1000 MW each
Construction licensing time	3.5 years
Unit construction time	6.5 years
Start-up dates	2027, 2030, 2033 & 2037
Reactor lifetime	60 years (including operation extension time).
Potential ownership model	Vendor/Owner-operator.
Potential contractual models	BOOT model (assumed “concession time”: 10 years).
Regulatory approach	Objectives-based.
Outsourcing policy	Standard.
National participation	Base Case Scenario - Maximum vendor contribution for the first national NPP project (thus, “minimum contribution from the country).

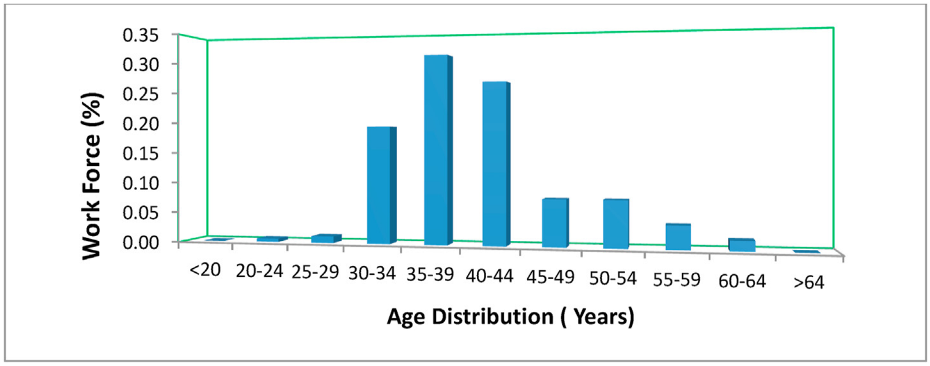

Fig. 3. Workforce distribution.

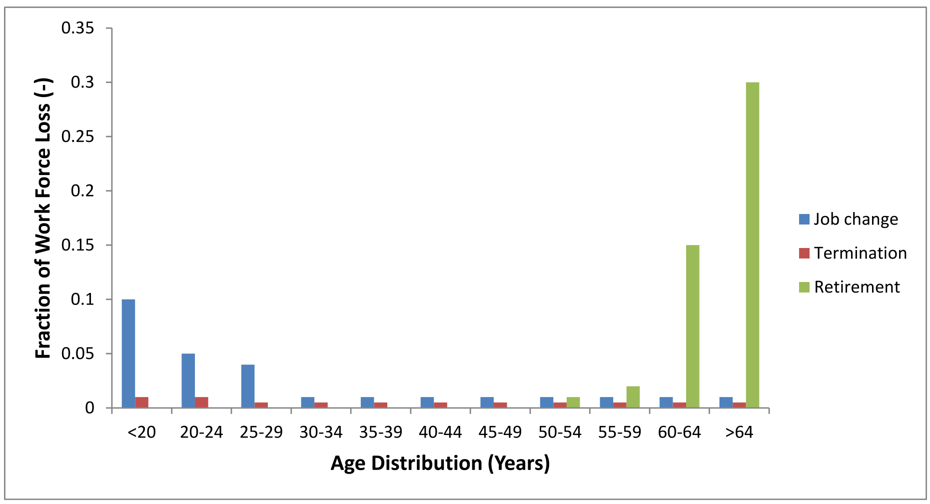

Fig. 4. Nuclear workforce attrition rate.

Fig. 5. Proposed sites for Nuclear Power Plants.

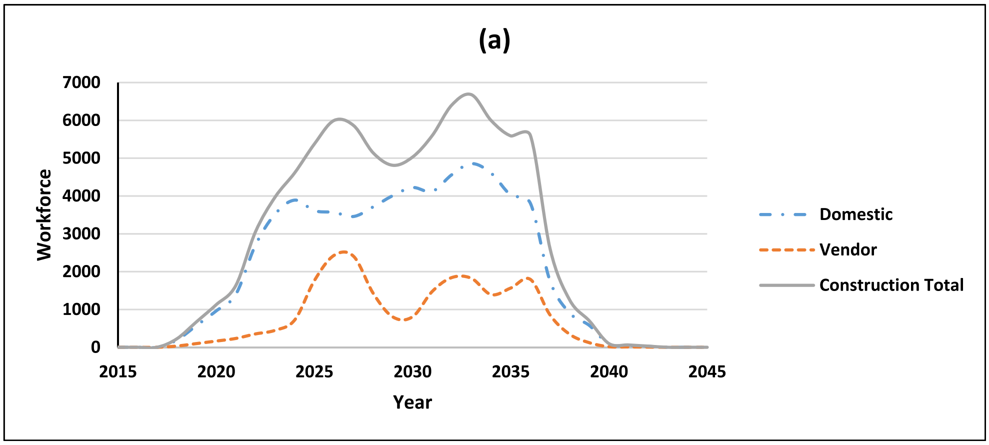

Fig. 6a. Proposed construction workforce.

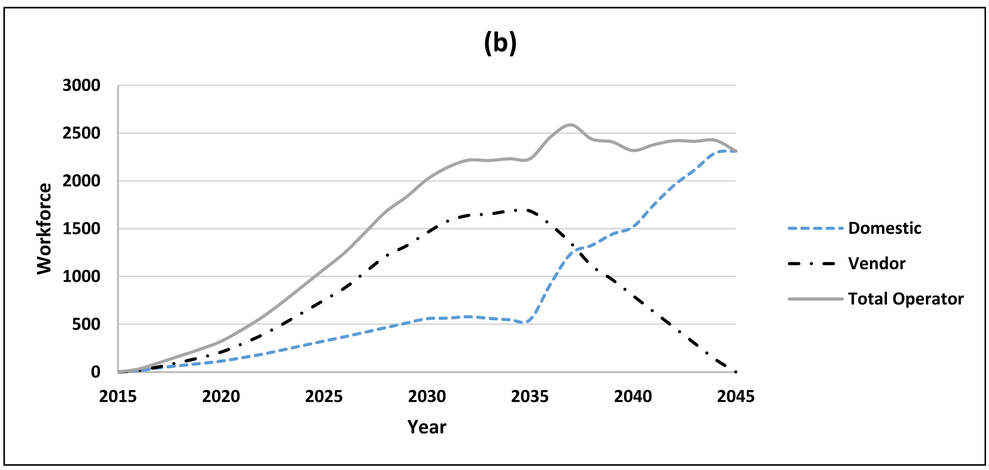

Fig. 6b. Proposed operator workforce.

should be derived from the vendor while the remainder should emanate from the domestic country. However, the peak total construction workforce steadily decreases to about 2606 personnel when the fourth unit is nearing completion around 2037.

Additionally, considering the operating staff (vendor and domestic), the results presented in Fig. 6b demonstrate a steady rise in the number of personnel required from about 1461 personnel during the commissioning of the first plant in 2027 to a peak of approximately 2586 staff in the year 2037 after the commissioning of the last unit in the same year. It is further demonstrated that at this peak workforce, the vendor contributes 52% (1346 personnel) to meet the workforce requirements while the remaining personnel (48%) are derived domestically. Furthermore, due to the BOOT contracting type and the proposed transition time of 10 years, the results show that the vendor contribution to the operating workforce gradually declines to zero in the year 2045.

Conversely, the regulatory body personnel as presented in Fig. 6c, depicts a slightly different story compared to the required operating staff and construction workforce. In this situation, the result shows that a maximum of about 228 people will be required during the construction phase which is about 6% higher than the required number in the Moroccan case study (see section 2.3). However, this number decreases to 154 personnel during the commissioning and steady operation of the fourth plant in 2037. The difference between the Moroccan case study and our study may be due to variances in the length of time required for the construction of each unit, construction licensing time and even the general terms of the contract.

Also, referring to Fig. 6a showing the proposed construction workforce, the total construction workforce fluctuates significantly as the nuclear power units commission comes online. This situation implies some construction workers may be relieved of their duties and then reappointed a few years down the line which is unsustainable for any NPP. To cater for this shortfall the construction workforce is split into civil craft workers and mechanical craft workers. While civil craft workers are responsible for planning, designing, and overseeing the construction and maintenance of building structures and infrastructure such as roads, railways, bridges, harbor, and water, and sewerage systems, mechanical craft construction workers are responsible for designing, developing, manufacturing, and installing new and modified components of the nuclear power plants units. A plot of these two construction workforce segments against time (in years for the commissioning of the plants) as displayed in Fig. 7 shows sustainable supply of the construction workers in the study time frame. It is thus pertinent and evident from Fig. 7 that the civil craft workers be deployed to the construction site before the mechanical craft workers.

An overall look at the three key organizations: construction, operator, and regulatory body, as depicted in Fig. 8, shows that a maximum of approximately 9045 personnel (73% construction, 24%

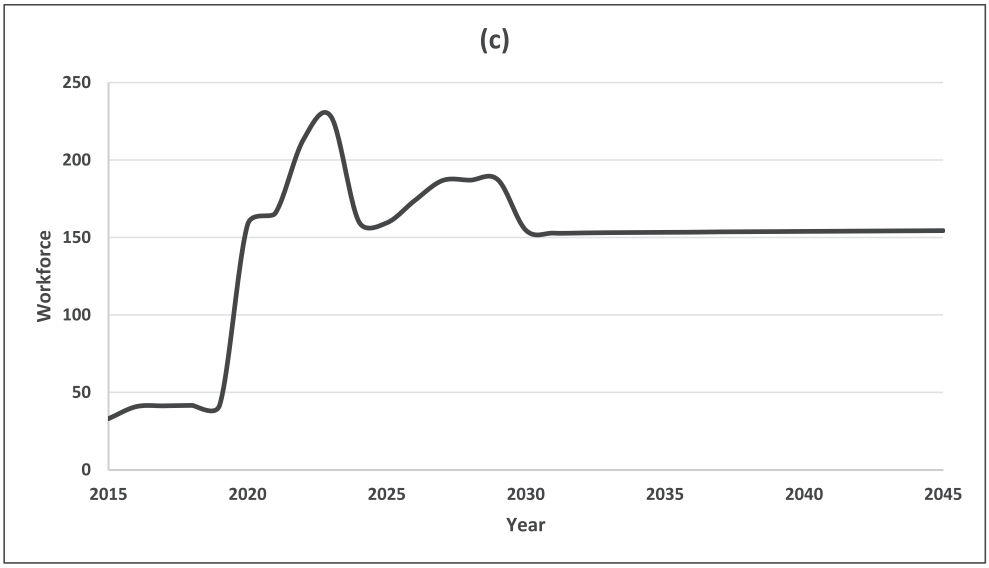

Fig. 6c. Proposed regulatory body workforce.

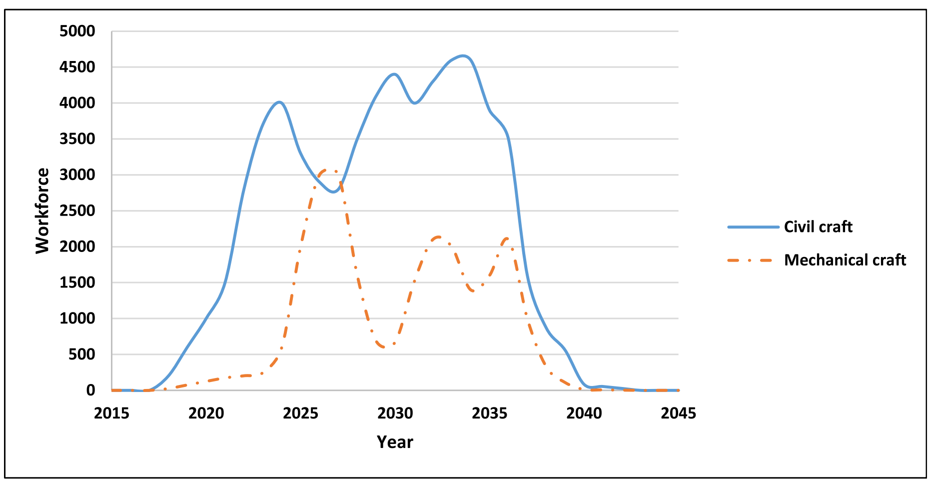

Fig. 7. Proposed civil craft and mechanical craft construction workforce.

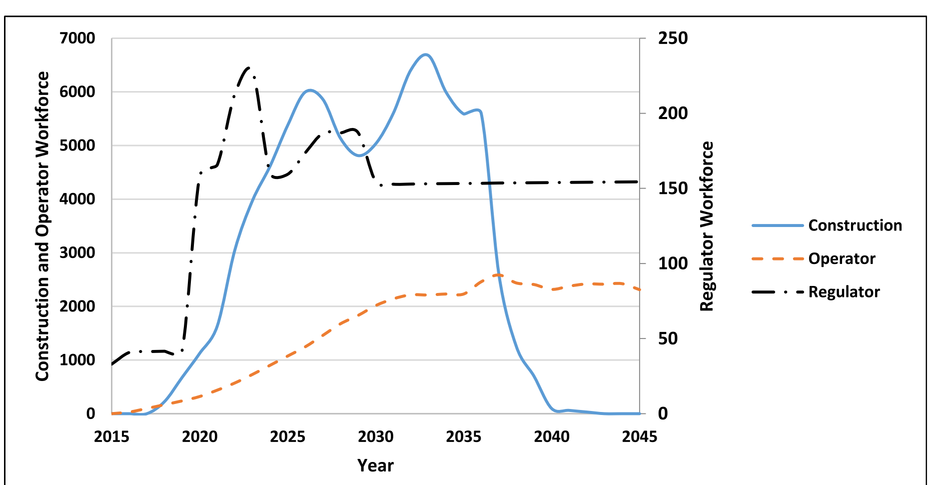

Fig. 8. Proposed overall workforce.

operators, and 3% regulators) should be engaged in the year 2033 just after the commissioning of the third NPP unit. The peak value of the combined workforce as presented in Fig. 8 is about 16% more than the Morocco case study discussed in section 2.3. Several reasons including the contractual agreement signed as stated earlier, technology used, and the level of skills available in a country could directly impact the results. Conversely, the workforce requirement fluctuates overtime but settles at approximately 2465 personnel requirements (94% operator and 6% regulator) in the year 2045.

In a similar manner, the NEPIO (which in this respect is NAEC i.e., the responsible organization tasked with leading and managing the direction and subsequent development of a national NPP) is only present in phases 1 and 2 of the NPP. During phase 1, the NEPIO acts as the nexus responsible for prescribing information to the government on the viability of proceeding with the NPP. If the government approves the viability of the NPP, the NEPIO may then be designated in Phase 2 to coordinate the development of the required NPP infrastructure among the various key organizations. Hence, as illustrated in Fig. 9, a maximum number of 50 workers should be engaged in the NEPIO in the year 2019 but this number should go to zero just before the year 2022 (i.e., at the end of phase 2 of the NPP). The result is similar to the work of the Moroccan case study (see section 2.3) following IAEA recommendations [31].

4.2. Domestic workforce requirements

A more in-depth look at each of the workforce skills required for the three key organizations (i.e. construction, owner/operator, and the regulatory body) as they relate to the domestic country shows that there are 12 disaggregated major craft skills required during the construction phase, following IAEA recommendations [32] including boilermakers, carpenters, electricians, iron workers, insulators, laborers, masons, millwrights, painters, pipefitters, sheet metal workers, and teamsters. The outcomes in Fig. 10 indicate that carpenters and laborers within the years 2019–2042 should account for at least 24% of the total labor force requirements. Moreover, the peak number of domestic construction craft workers potentially required within the time frame considered (i.e., 2015 to 2045) is approximately 4856 personnel occurring in the year 2033. A breakdown of the workforce within the same year (i.e., 2033) portrays the following makeup: boilermakers (0.30%), carpenters (27.29%), electricians (6.67%), iron workers (1.33%), insulators (5.46%), laborers (27.29%), masons (1.36%), millwrights (7.37%), painters (5.46%), pipefitters (3.15%), sheet metal workers (8.19%), and teamsters (6.14%). It may be observed that carpenters and laborers constitute over 50% of the domestic construction craft workers. This is attributed to the domestic country providing most of the less skilled crafts in contrast to the vendor providing such scarce skills as boiler making.

On the other hand, the regulatory body workforce requires skills in the following areas: technical, safeguards, legal, science, security, and others (mostly administrative) [33]. The model illustrates in Fig. 11, that technical staff should be recruited the most. Moreover, as stated earlier in section 4.1, the maximum number of regulatory workers needed is about 228 people occurring in the year 2023 and requiring a breakdown of the following skill sets: technical (17.85%), safeguards (17.15%), legal (16.49%), science (16.49%), security (15.83%) and others (16.18%).

However, it is pertinent to note that domestic workers' skill requirements for the operating organization as illustrated in Fig. 12 may be delineated into three categories according to the level of education obtained: professionals (with minimum Bachelor's degree), technicians (with Higher National Diploma), and craftsmen (with at least a secondary school certificate). From Fig. 12, considering the year 2027 when the first NPP unit is proposed to be commissioned to the year 2037 for the fourth unit commissioning, the professional workforce should consistently account for at least 45% of the domestic operating workforce. The craftsmen should account for a minimum of 36% of the workforce while the technicians take up the remaining workforce requirements.

Lastly, the age distribution for the operator and regulator organizations at the end of the model period (i.e., 2045) as displayed in Fig. 13 differs from the proposed age distribution presented in section 3.2 regarding the key assumptions (see Fig. 3). While the median age range for the operator and regulator occurs between 45 – 49 and 40–44 age range respectively, the median age in the proposed age distribution (see Fig. 3) is in the 35–39 year range.

5. Skill set needed in Nigeria

A preliminary analysis of typical workforce requirements for the sustained operation of nuclear power plants (up to commercial operation) is illustrated in Table 3 (and presented in the supplementary material “staffing”). Table 3 indicates that about 21% of the workforce education and training programs should be tailored to developing management and support services related skills and so

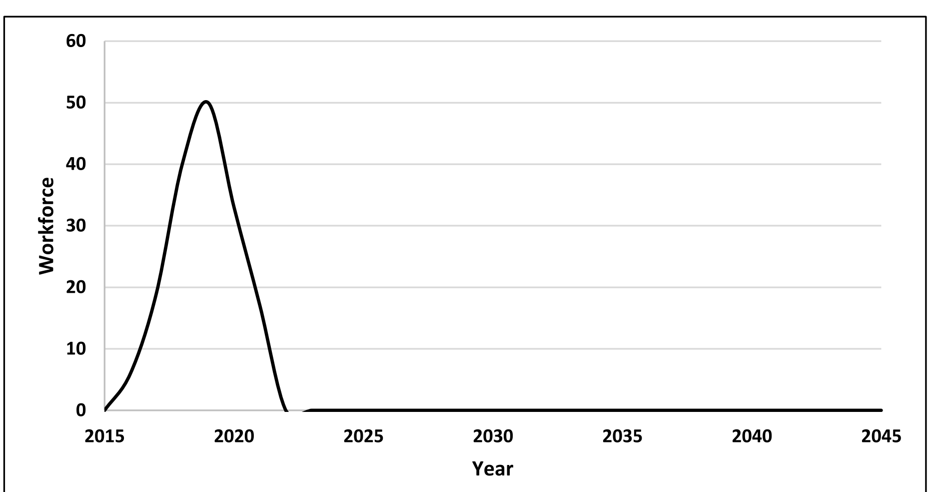

Fig. 9. Proposed NEPIO workforce.

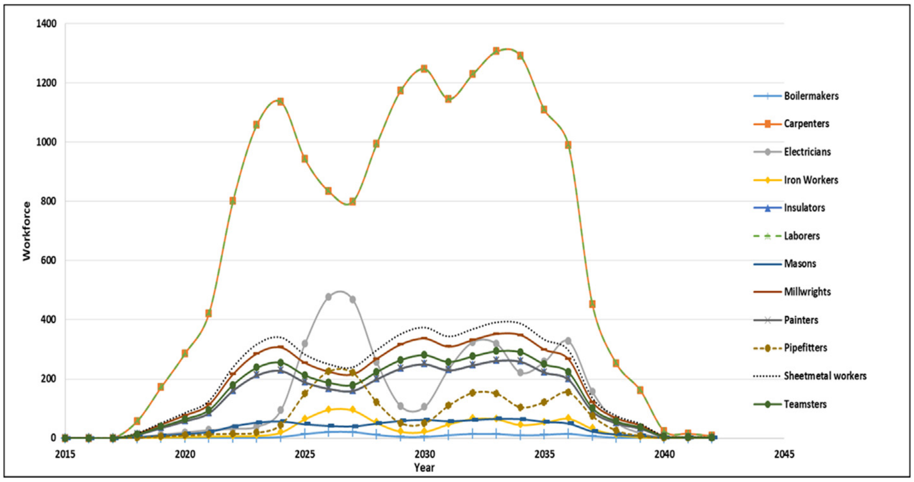

Fig. 10. Proposed Craft skills during Construction.

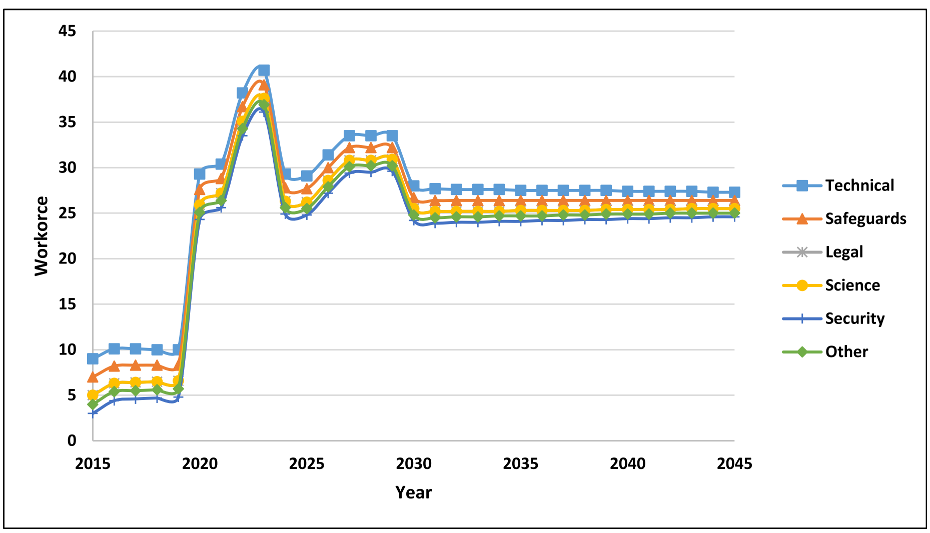

Fig. 11. Proposed regulatory body skills.

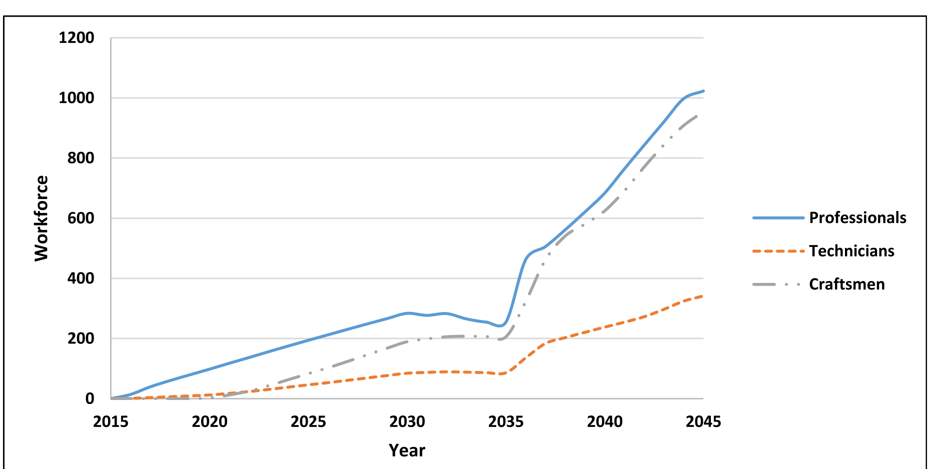

Fig. 12. Proposed operating organization Educational Qualification level.

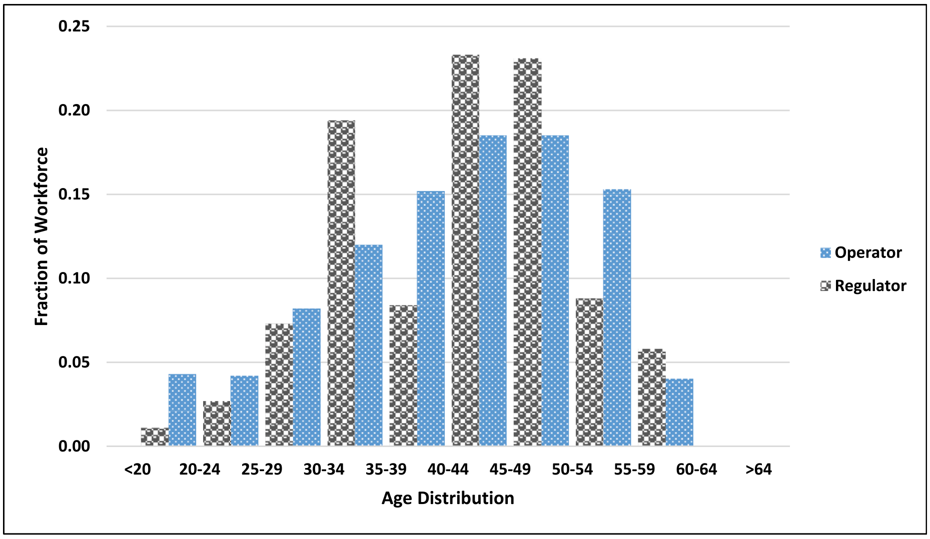

Fig. 13. Age distribution of operator and regulatory staff in the year 2045.

on. However, for Nigeria to develop all the critical skills, a three-prong approach as stipulated in the Human Resource Development Plan for Nigeria's nuclear energy program [2] is being explored as follows:

5.1. Vendor support for education and training

As stated in the introduction (Section 1), Nigeria has signed two project development agreements (PDAs) with the Russian Federation in different aspects of NPP development. As a backdrop of the agreements, the Russian Federation is providing fully-sponsored scholarship schemes for graduate education and technical programs in nuclear science and engineering to bolster some of the NPP needed skills. The scheme commenced in 2016 and is showing an exponential uptake of Nigerians.

5.2. International support for education and training

Nigeria also has several collaborations and partnership agreements with the IAEA, the African Regional Cooperative Agreement for Research and Development in Nuclear Science and Technology (AFRA), World Institute for Nuclear Security (WINS), Partnership for Nuclear Security (PNS), and the World Nuclear University (WNU) in the form of multilateral cooperation. Moreover, there are also bilateral agreements with South Korea and China for postgraduate-level education. Also, through the IAEA and NPP operating countries, fellowships and scientific visits are being granted as part of the sustained training and education of the NPP workforce.

5.3. National support for education and training

In the quest to domesticate the education and training of nuclear professionals, Nigeria is developing infrastructure and capacity through Memoranda of Understanding (MOU) with eleven local universities and polytechnics. The partnerships are to accelerate the implementation of nuclear science and engineering programs at the diploma and graduate levels. Furthermore, Nigeria through NAEC is exploring potential collaboration with such local institutions as the National Power Training Institute of Nigeria (NAPTIN) and the Petroleum Training Institute for the training of technicians and craftsmen.

Moreover, Nigeria has six existing education and training centers located in its six geopolitical zones with various infrastructures and facilities such as two generic computer-based NPP simulators for training, a 340 kCi Co-60 Gamma Irradiation Facility, a 30 kW miniature neutron source reactor (MNSR), a 1.7 MeV Tandem Accelerator [34]. However, there will be a need to acquire more infrastructure as the NPP develops.

Alternatively, there are several regulations developed not only to guide the nuclear safety and security culture of current and future nuclear professionals but best practices as prescribed by the IAEA. Some of these regulations (see below) developed by the Nigerian Nuclear Regulatory Authority (NNRA) [35] form the fulcrum of Nigeria's nuclear power program.

i. Nigerian Safety and Security Regulation
ii. Nigerian Radioactive Waste Management Regulation
iii. Nigerian Nuclear Safeguards Regulations, 2021
iv. Nigerian Safety Regulations for Licensing of Sites for Nuclear Power Plants, 2021
v. Nigerian Uranium Exploration, Mining and Processing regulations, 2021
vi. Nigerian Physical Protection of Nuclear Material and Nuclear Facilities Regulations, 2021

Furthermore, a safety regulation on the design and construction of NPPs has been submitted to the Nigerian government for gazette while guidelines and regulations for the education and training of personnel including operators of nuclear facilities are almost finalized. In addition, a National Nuclear Security Committee headed by the National Security Adviser (NSA) of Nigeria, comprising the Regulator, NEPIO, Academia, and other relevant stakeholders has been established to ensure the Nuclear Security Regime developed is implemented by and complied with by all staff and professionals.

The NNRA has provided best practices in developing regulations by providing drafts of regulations to the operator and NEPIO for comments and inputs before finalization. Furthermore, another best practice employed is the continuous and up-to-date regulations the NNRA provides to the Nigerian nuclear industry operators. The regulatory body has over 10 years of experience in maintaining good practices of regulating the safe operation of the Nigerian Research Reactor-1 (NIIR-1), hence, will be extended to the regulation of nuclear power plants.

Table 3
Required process area and job function.

	Plant Design Selection	Site Selection	Plant Licensing	Construction	Start-up/Testing	Commercial Operation
Operations						
Chemistry	–	–	15	15	15	15
Environment		4	4	4	4	4
Operations	93		93	93	93	93
Operations Support					10	10
Subtotal	**93**	**4**	**112**	**112**	**122**	**122**
Maintenance						
Maintenance Construction	140			140	140	140
Maintenance/Constr. Support				20	20	20
Subtotal	**140**			**160**	**160**	**160**
Supply Chain						
Contracts Material Management	1	1		1	1	1
				3	3	3
Purchasing/Contracts	4	4		4	4	4
Warehouse				9	9	9
Subtotal	**5**	**5**		**17**	**17**	**17**
Management and Support Serv.						
Admin/Clerical	33	33	33	33	33	33
Budget/Accounting	7	7	7	7	7	7
Communications		2	2	2	2	2
Document Control	11	11	11	11	11	11
Facilities				31	31	31
Human Resources			6	6	6	6
Info Systems			17	17	17	17
Management	30	30	30	30	30	30
Management Support				2	2	2
Training			40	40	40	40
Subtotal	**81**	**83**	**146**	**179**	**179**	**179**
Work Management						
Outage Management						15
Project Management					12	12
Scheduling				15	15	15
Subtotal				**15**	**27**	**42**
Engineering & Tech. Support						
Computer Engineering	5		5	5	5	5
Design Engineering	42	42	42	42	42	42
Design/Drafting			8	8	8	8
Nuclear Fuels	7		7	7	7	7
Procurement Engineering					6	6
Quality Control				7	7	7
Reactor Engineering	5		5	5	5	5
System/Field Engineering	28		28	28	28	28
Technical Engineer	22	22	22	22	22	22
Subtotal	**109**	**64**	**117**	**124**	**130**	**130**
Safety						
Emergency Planning		6	6	6	6	6
Fire Protection		2	2	2	2	2
Safety/health			3	3	3	3
Security		123	123	123	123	123
Subtotal		**131**	**134**	**134**	**134**	**134**
Radiation Protection						
ALARA/Rad Engr	6		6	6	6	6
Radwaste/Decon			17		17	17
HP Applied				14	14	14
HP Support			9	9	9	9
Subtotal	**6**		**32**	**29**	**46**	**46**
Regulatory Compliance						
Licensing	7	7	7	7	7	7
Nuclear Safety Review			6	6	6	6
Quality Assurance	11	11	11	11	11	11
Subtotal	18	18	24	24	24	24
TOTAL	**352**	**305**	**565**	**794**	**839**	**850**
Percentage (%)	**42**	**36**	**67**	**93**	**99**	**100**

6. Conclusion

The IAEA milestone approach lists human resource development as one of the nineteen (19) infrastructure issues to be addressed before a nuclear power plant (NPP) can be introduced into a country's energy mix. The results of this study showed that for the timeframe considered (from 2015 to 2045) while reflecting on the three key organizations: construction, owner/operator, and regulatory body, the first 15 years (2015–2030) illustrate the dominance of construction workforce in the program. Moreover, the number of construction workers peaked at approximately 73% (6678 personnel) of the combined workforce just before the commissioning of the third nuclear power plant in 2033. However, the results showed a steady rise in the requirement for owner/operators which plateaus at about 2400 personnel a couple of years after the fourth (4th) nuclear power plant should have been commissioned. The regulatory workforce steadily contributes a little less than 10% of the combined key nuclear organizations workforce requirements.

There are many more merits of the NPHR modeling tool for planning and assessing nuclear workforce requirements. However, it is hoped that in future work scenario analyses will be explored. For instance, it will be interesting to assess the effects of varying the vendor contribution to the workforce, or even the vendor contracting type approach (turnkey or build-own-operate or a hybrid of vendor contracting types). Moreover, this study provides the base case scenario of the open-source policy statements of Nigeria whereby a gap analysis will be carried out in future work. Lastly, it is hoped that an economic perspective to the NPHR model will be integrated since the payment of salaries accounts for at least 25% of the operating and maintenance cost of nuclear power plants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.net.2021.08.019.

References

[1] J.M. Egieya, L. Cucek, K. Zirngast, A.J. Isafiade, B. Pahor, Z. Kravanja, Synthesis of Biogas Supply Networks using various biomass and manure types, Comput. Chem. Eng. 122 (2018) 129–151, https://doi.org/10.1016/j.compchemeng.2018.06.022.

[2] NAEC, History of the Nigeria Atomic Energy Commission, 2021. Retrieved from, https://www.nigatom.org.ng/history-of-the-commission/. (Accessed 3 March 2021). Accessed.

[3] Worldometer, Population of Nigeria and growth rate, 2021. https://www.worldometers.info/world-population/nigeria-population/. (Accessed 4 March 2021). Accessed.

[4] A.A. Ujile, Evaluating water-energy-food nexus for sustainability of healthy living in Nigeria 2, Nigerian Society of Chemical Engineers Publication, 2020, pp. 1–18, 1.

[5] P. Burkhardt, Russia Signs Agreement with Nigeria for Nuclear Power Plant, 2017. Retrieved from, https://www.bloomberg.com/news/articles/2017-10-30/russia-signs-agreement-with-nigeria-to-build-nuclear-power-plant. (Accessed 20 March 2021). Accessed.

[6] IAEA, Workforce planning for new nuclear power programs, IAEA nuclear energy series No. NG-T-3.10, 2011.

[7] IAEA, Milestones in the Development of a National Infrastructure for Nuclear Power, Nuclear Energy Series No. NG-G-3.1 (Rev. 1), IAEA, Vienna, 2015.

[8] F. Alam, R. Sarkar, H. Chowdhury, Nuclear power plants in emerging economies and human resource development: a review, Energy Procedia 160 (2019) 3–10, https://doi.org/10.1016/j.egypro.2019.02.111.

[9] S.N. Mishra, D.R. Lama, Y. Pal, Human resource predictive Analytics (HRPA) for HR management in organizations, International Journal of Scientific & Technology Research 5 (2016) 33–35. ISSN 2277-8616.

[10] J.O. Samwel, Human resource planning as an important practice to anticipate future human resource requirements of the organization – literature review, International Journal of Research in Business Studies and Management 5 (2018) 24–30. ISSN 2394-5931;
[a] B.M. Lillehol, K. Lønning, M.H. Andersen, Exploring nurse managers' perception of using the RAFAELA system as a management tool in a Norwegian hospital setting, Nursing Open 5 (1) (2018) 77–83, https://doi.org/10.1002/nop2.115.

[11] B.O. Akinnuli, R.K. Apalowo, Regression analysis based effective manpower planning methodology: a case study, Journal of Engineering Research and Reports 1 (4) (2018) 1–12, https://doi.org/10.9734/jerr/2018/v1i49878.

[12] V. O Ezugwu, S. Ologun, Markov chain: a Predictive model for manpower planning, J. Appl. Sci. Environ. Manag. 21 (3) (2017) 557–565.

[13] S. Tachapon, Dynamic Human Resource Predictive Model for Complex Organizations. Master's Thesis, University of Tennessee, 2011. Retrieved from Available, https://trace.tennessee.edu/utk_gradthes/1019:08-20. (Accessed 4 April 2021).

[14] M.K. Onifade, O.J. Afolabi, O.A. Oroye, Modeling the manpower planning of Nigerian Port operations, J. Academic Res. Econ. 12 (3) (2020) 494–508.

[15] H. Kwon, J. Byung, L. Eui-Jin, Y. Byung-Hoon, Prediction of human resource supply/demand in nuclear industry using Markov chain model and job coefficient, in: Transactions of Korean Nuclear Society Autumn Meeting, Gyeongin, Korea, November 2-3, 2006.

[16] F. Alam, R. Sarkar, A.R. Paul, A. Aldiab, Nuclear power plants and human resources development in South asia, in: Proceedings of the 7th International Conference on Advances in Energy Research, Springer, Singapore, 2020, pp. 1603–1613.

[17] P.H. K Ho, Forecasting construction manpower demand by gray model, J. Construct. Eng. Manag. 136 (12) (2010) 1299–1305, https://doi.org/10.1061/(ASCE)CO.1943-7862.0000238.

[18] K. Fujiwara, Japan nuclear human resource development network, Energy Procedia 127 (2017) 35–43, https://doi.org/10.4314/jasem.v21i3.17.

[19] IAEA, Evaluation of Human Resource Needs for a New Nuclear Power Plant: Armenian Case Study, IAEA-TECDOC-1656, IAEA, Vienna, 2011b.

[20] M.S. Akbar, Y.N. Jolly, An approach of manpower development for new nuclear power programs: a case study of Bangladesh, Int. J. of Nuclear Governance 4 (2013) 19–40, https://doi.org/10.1504/IJNGEE.2013.053812.

[21] UAE, UAE national report for the 7th review meeting of the convention on nuclear safety, 2021. Retrieved from, https://www-ns.iaea.org/downloads/ni/safety_convention/7th-review-meeting/17-08671e_cns7rm2017_08.pdf. (Accessed 4 April 2021).

[22] WNA, Nuclear Power in the United Arab Emirates Nuclear Power in the United Arab Emirates, 2021. Retrieved from, https://www.world-nuclear.org/information-library/country-profiles/countries-t-z/united-arab-emirates.aspx. (Accessed 4 March 2021).

[23] IAEA, Experience on HRD analysis and use of the NPHR modeling tool: country case studies, 2021. Unpublished report.

[24] A. Gurgen, S. Ergun, A sample application of nuclear power human resources model, BgNS Transactions 21 (57) (2016) 54. ISSN 1310-8727.

[25] E. Dyck, A. Evrensel, IAEA helps nuclear newcomers with workforce plan, IAEA Department of Energy (2015). Retrieved from, https://www.iaea.org/newscenter/news/iaea-helps-nuclear-newcomers-workforce-planning-tool. (Accessed 24 April 2021). Accessed.

[26] Isee systems, Stella and ithink version 1.3.1, Retrieved from, https://www.

iseesystems.com/resources/help/v1-3/default.htm, 2017. (Accessed 3 March 2021). Accessed.
[27] J.D. Sterman, Business Dynamics: Systems Thinking and Modeling for a Complex World, sixth ed., McGraw Hill India, New Delhi, 2000.
[28] G.E. invest, Nigeria energy sector, Retrieved from, https://www.get-invest.eu/market-information/nigeria/energy-sector/, 2021. (Accessed 24 April 2021). Accessed.
[29] IAEA-PRIS, Database on nuclear power reactors, Retrieved from, https://pris.iaea.org/pris/, 2021. (Accessed 22 March 2020). Accessed.
[30] IAEA, Responsibilities and capabilities of a nuclear energy program implementing organization (NEPIO), IAEA Nuclear Energy Series No. NG-T-3.6 (2009).
[31] Department of Energy DOE, NP2010 nuclear power plant construction infrastructure assessment. MPR-2776, 2005.
[32] IAEA, Preparation of a feasibility study for new nuclear power projects, IAEA Nuclear Energy Series No. NG-T.3.3 (2014).
[33] D.R.E. Ewim, S.S. Oyewobi, M.O. Dioha, C.E. Daraojimba, S.O. Oyakhire, Z. Huan, Exploring the perception of Nigerians towards nuclear power generation, African J. Sci. Technol. Innov. Dev. (2021), https://doi.org/10.1080/20421338.2021.1930848.
[34] NNRA, Nigerian Nuclear Regulatory Authority Regulations, 2021. Retrieved from, https://nnra.gov.ng/nnra/resources?d=1&category=nnra_regulations. (Accessed 8 August 2021).